# AI-Powered Sales Analytics System Using Natural Language Queries

**Project Report**

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## 1. Introduction

Businesses generate large volumes of data every day. Extracting meaningful insights from this data typically requires knowledge of database queries and analytical tools. Non-technical users often find it difficult to retrieve insights directly from databases.

This project presents an AI-powered analytics system that allows users to ask questions in natural language. The system automatically converts these questions into SQL queries, retrieves data from the database, and generates insights along with visualizations.

The workflow automation and orchestration were implemented using n8n, while the dataset was stored and queried using PostgreSQL.

## 2. Objective of the Project

- Allow users to query business data using natural language.
- Automatically generate SQL queries from user prompts.
- Retrieve and analyze data from a database.
- Generate meaningful insights and charts automatically.
- Demonstrate how workflow automation and AI simplify business analytics.

## 3. Problem Statement

Traditional data analysis requires knowledge of SQL queries and database structures. Business users often depend on technical teams to generate reports or insights from data.

This project addresses this challenge by creating an automated system where users can ask questions in natural language and the system will automatically generate SQL queries, retrieve data, and present the results in an easy-to-understand format.

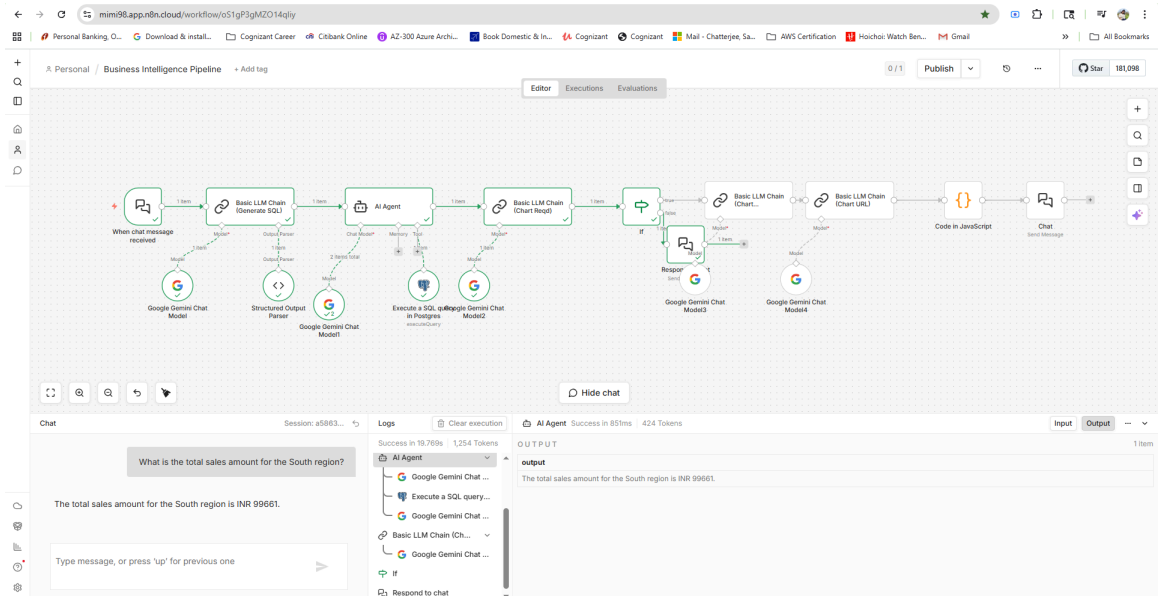
## 4. Tools and Technologies Used

| Tool / Technology | Purpose |
|---|---|
| n8n | Workflow automation and orchestration |
| PostgreSQL | Database management system |

| | |
|---|---|
| SQL | Data querying |
| AI Language Model | Natural language understanding and SQL generation |
| Chart Visualization | Graphical representation of data |

## 5. System Architecture

1. User enters a question in natural language.
2. The AI model analyzes the prompt.
3. A SQL query is generated automatically.
4. The query is executed in the PostgreSQL database.
5. Results are returned to the workflow.
6. The system analyzes the data.
7. Charts and insights are generated.
8. Final results are displayed to the user.




## 6. Dataset Description

The project uses a sales dataset containing information about customer purchases across regions.

| Column Name | Description |
|---|---|

| | |
|---|---|
| Region | Geographic region of sales |
| Customer_Type | Indicates whether the customer is New or Returning |
| Product_Category | Category of the product |
| Sales_Amount | Total revenue generated |
| Sales_Representative | Sales representative responsible for the transaction |

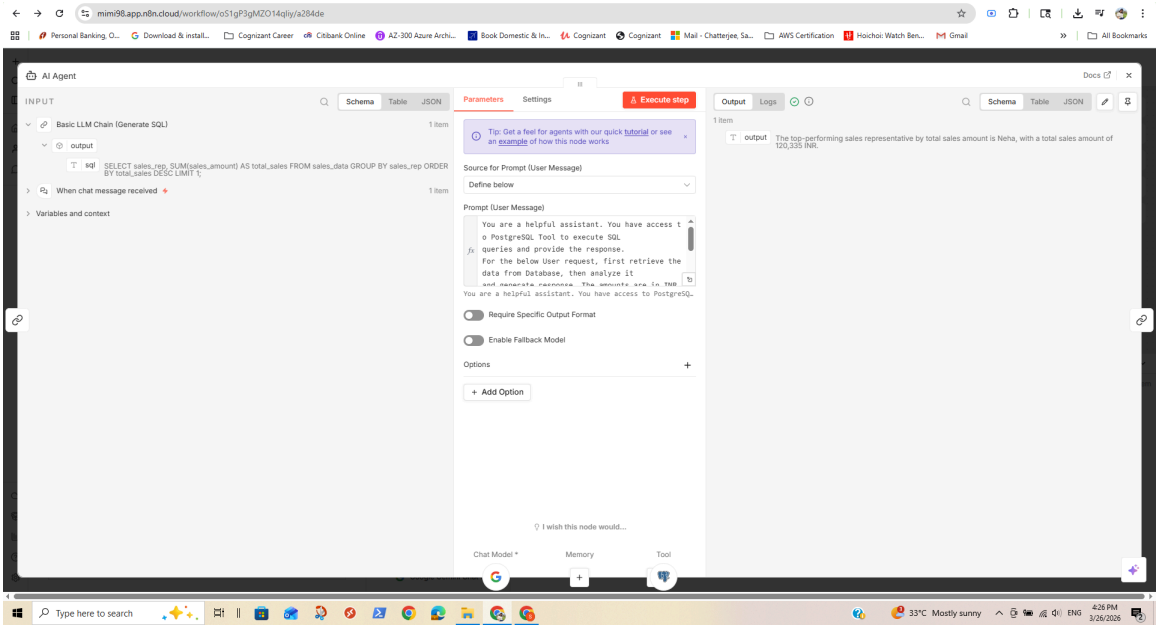
## 7. Workflow Implementation

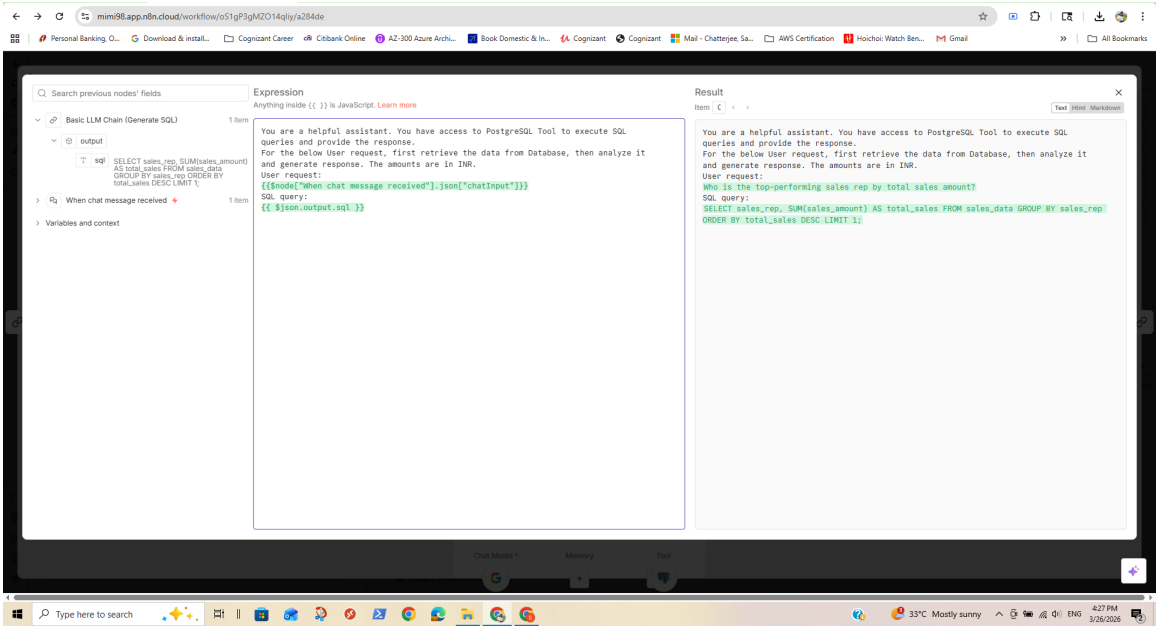
Step 1 – User Query Input

The user enters a natural language question such as: Compare total sales for New vs Returning customers across regions.

Step 2 – SQL Query Generation

The AI model interprets the question and generates the SQL query automatically.






Step 3 – Query Execution

The generated SQL query is executed in the PostgreSQL database to retrieve relevant data.

mimi98.app.n8n.cloud/workflow/oS1gP3gMZO14qliy/0da952

Basic LLM Chain (Generate SQL)

INPUT

When chat message received

action: sendMessage

sessionId: a58634b77d78490f810e0632cb8be255

chatInput: Who is the top-performing sales rep by total sales amount?

Variables and context

Parameters | Settings | Execute step

Save time with an example of how this node works

Source for Prompt (User Message): Define below

Prompt (User Message)

```
You are a PostgreSQL Expert. Convert the use
r's question into an optimized SQL query.

The sales data is stored in a PostgreSQL table
called `sales_data` with the following columns
and meanings:
```

Require Specific Output Format

Enable Fallback Model

Chat Messages (if Using a Chat Model)

+ Add prompt

Batch Processing

+ Add Batch Processing Option

Connect an output parser on the canvas to specify the output format you require

I wish this node would...

Model * | Output Parser

Output | Logs

output

sql: SELECT sales_rep, SUM(sales_amount) AS total_sales FROM sales_data GROUP BY sales_rep ORDER BY total_sales DESC LIMIT 1;

Type here to search | 33°C Mostly sunny | ENG | 4:18 PM 3/26/2026

mimi98.app.n8n.cloud/workflow/oS1gP3gMZO14qliy/0da952

Search previous nodes' fields

When chat message received

action: sendMessage

sessionId: a58634b77d78490f810e0632cb8be255

chatInput: Who is the top-performing sales rep by total sales amount?

Variables and context

Expression

Anything inside {{ }} is JavaScript. Learn more

```
You are a PostgreSQL Expert. Convert the user's question into an optimized SQL query.

The sales data is stored in a PostgreSQL table called `sales_data` with the following
columns and meanings:

row_id (INTEGER) – unique identifier for each row
product_id (INTEGER) – unique identifier for each product
sale_date (DATE) – date on which the sale occurred
sales_rep (TEXT) – name of the sales representative
region (TEXT) – geographic region (North, South, East, West)
sales_amount (NUMERIC) – total revenue generated from the sale
quantity_sold (INTEGER) – number of units sold
product_category (TEXT) – category of the product (Furniture, Electronics, Food,
Clothing)
unit_cost (NUMERIC) – cost per unit of the product
unit_price (NUMERIC) – selling price per unit
customer_Type (TEXT) – New or Returning customer
discount (NUMERIC) – discount applied on the sale
payment_Method (TEXT) – Cash, Credit Card, Bank Transfer
sales_channel (TEXT) – Online or Retail
region_and_sales_rep (TEXT)

User Question:
{{$json["chatInput"]}}

Return only a valid SQL query in below json format, no explanation:

{
"sql": "SELECT * FROM sales_data LIMIT 5;"
}
SQL Generation Rules:

1. If the user asks for comparisons across categories (e.g., region, product
category, customer type, sales rep), include those columns in the SELECT clause.

2. When multiple categories are involved, include all relevant fields in the GROUP BY
clause.
```

Result

Item C | Text | Html | Markdown

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User Question:
Who is the top-performing sales rep by total sales amount?

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{
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1. If the user asks for comparisons across categories (e.g., region, product
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Type here to search | Air: Poor | ENG | 4:18 PM 3/26/2026

mimi98.app.n8n.cloud/workflow/oS1gP3gMZO14qliy/0da952

Search previous nodes' fields

When chat message received — 1 item

- action: sendMessage
- sessionId: a58634b77d78490f810e0632cb8be255
- chatInput: Who is the top-performing sales rep by total sales amount?

Variables and context

**Expression**

Anything inside {{ }} is JavaScript. Learn more

```
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2. When multiple categories are involved, include all relevant fields in the GROUP BY
clause.

3. If the user mentions phrases like:
   - "across regions"
   - "by region"
   - "per region"
   - "across categories"

   then include the region column in both SELECT and GROUP BY.

4. Always ensure the query returns enough data for meaningful comparison or
visualization.
```

**Result**

Item | C

Text | Html | Markdown

```
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Model * | Output Parser

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mimi98.app.n8n.cloud/workflow/oS1gP3gMZO14qliy/2d8b57

**Execute a SQL query in Postgres** — Docs

INPUT — Mapping | From AI — Schema | Table | JSON

1 item

No fields - item(s) exist, but they're empty

Parameters | Settings — Execute step

Credential: Postgres account

Tool Description: Set Automatically

Operation: Execute Query

Query:

```
1  {{ $json.output.sql }}
```

Consider using query parameters to prevent SQL injection attacks. Add them in the options below

Options

+ Add option

I wish this node would...

OUTPUT — Schema | Table | JSON

No parameters are set up to be filled by AI. Click on the button next to a parameter to allow AI to set its value.

8 items

- customer_type: New
- region: East
- total_sales: 52281

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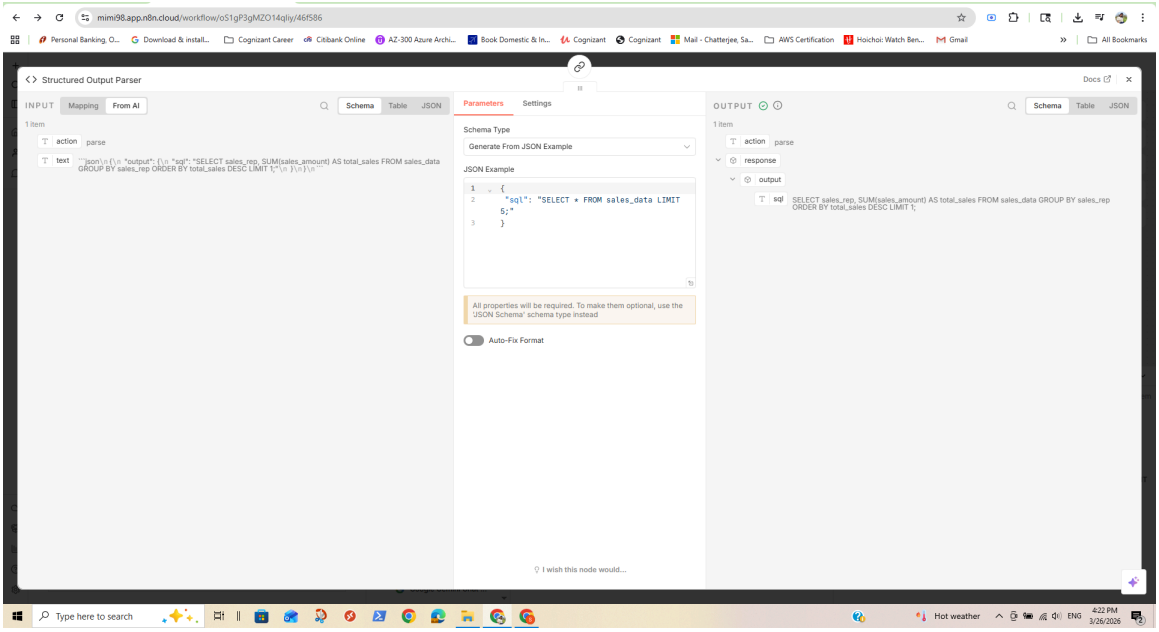
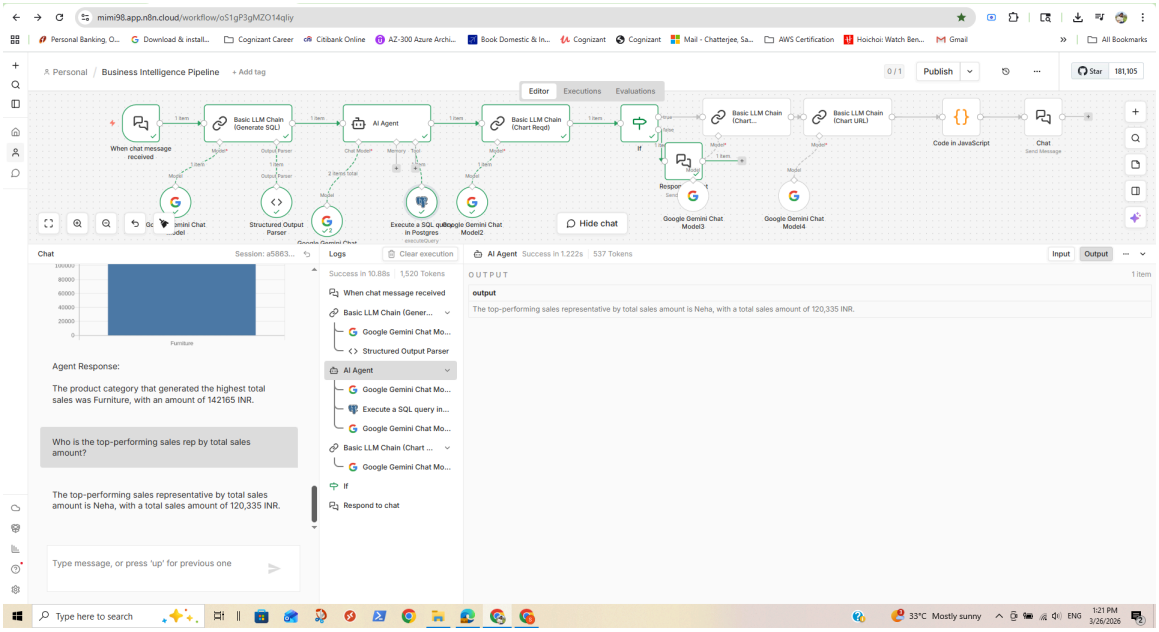



## Step 4 – Data Processing

The retrieved data is analyzed to extract meaningful insights.

## Step 5 – Chart Generation

The workflow generates charts and visualizations to represent the results clearly.



## 8. Sample Output

- East Region – New Customer Sales: 52,281 INR; Returning Customer Sales: 81,960 INR.
- North Region – New Customer Sales: 81,101 INR; Returning Customer Sales: 45,849 INR.
- South Region – Returning customers generated higher revenue compared to new customers.
- West Region – Returning customers showed the highest sales contribution.

## 9. Key Features of the System

- Natural language query support
- Automatic SQL query generation
- Database integration with PostgreSQL
- Automated data analysis
- Dynamic chart generation
- User-friendly analytics workflow

## 10. Challenges Faced

- Handling incorrect SQL generation from AI responses.
- Ensuring output format compatibility with chart generation.
- Debugging workflow nodes when data formats changed.
- Managing structured JSON outputs for database execution.

## 11. Conclusion

This project demonstrates how AI and workflow automation can simplify business data analysis. By integrating natural language processing with database querying and automated visualization, the system enables users to obtain insights without requiring technical expertise.

## 12. Future Scope

- Voice-based query support
- Real-time dashboard integration
- Integration with BI tools
- Support for larger and real-time datasets